

Question-9.4.35

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Question

A train travels a distance of 480 km at a uniform speed. If the speed had been $8\frac{\text{km}}{\text{h}}$ less, then it would have taken 3 hours more to cover the same distance. We need to find the speed of the train.

Solution

Let the unknown speed be x .

$$t = 480 \left(\frac{\frac{1}{x-8}}{\frac{1}{x}} \right) \quad (1)$$

$$\mathbf{A}^T \mathbf{X} = 3, \text{ where } \mathbf{A} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} \frac{480}{x-8} \\ \frac{480}{x} \end{pmatrix} \quad (2)$$

Multiply both sides by $x(x-8)$:

$$480x - 480(x-8) = 3x(x-8) \quad (3)$$

$$480x - 480x + 3840 = 3x^2 - 24x \quad (4)$$

$$3x^2 - 24x - 3840 = 0 \quad (5)$$

$$\begin{pmatrix} x^2 & x & 1 \end{pmatrix} \begin{pmatrix} 3 \\ -24 \\ -3840 \end{pmatrix} = 0 \quad (6)$$

Solution

Discriminant:

$$D = (-24)^2 - 4 \times 3 \times (-3840) = 46656 \quad (7)$$

$$x = \frac{24 \pm \sqrt{46656}}{6} = \frac{24 \pm 216}{6} \quad (8)$$

Positive root:

$$x = \frac{24 + 216}{6} = 40 \quad (9)$$

Therefore, train speed = $40 \frac{km}{h}$

```
// code.c
#include <stdio.h>

void get_roots(double *root1, double *root2) {
    double a = 3.0;
    double b = -24.0;
    double c = -3840.0;
    double D = b*b - 4*a*c;
    if (D < 0) {
        *root1 = 0.0;
        *root2 = 0.0;
    } else {
        double sqrtD = sqrt(D);
        *root1 = (-b + sqrtD) / (2*a);
        *root2 = (-b - sqrtD) / (2*a);
    }
}
```

```
# Code by GVV Sharma  
# Modified for Problem Solution  
# Released under GNU GPL  
# Calculating area enclosed between curves  
# call.py  
import ctypes  
import numpy as np  
  
lib = ctypes.CDLL('./code.so')
```

```
get_roots = lib.get_roots
get_roots.argtypes = [
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_double)
]
get_roots.restype = None

def get_quadratic_roots():
    root1 = ctypes.c_double()
    root2 = ctypes.c_double()
    get_roots(ctypes.byref(root1), ctypes.byref(root2))
    return root1.value, root2.value
```